

A small supervision budget barely improves weak-to-strong generalization

On GPT-2 / BoolQ, mixing a ground-truth budget into the weak labels helps only modestly — and *where and how* it is spent show little effect.

A consistent account: the student largely reproduces the teacher's errors, and ground truth corrects mainly the examples it directly labels — so the effective lever is *volume*, not placement or combination.

GPT-2 family · BoolQ + SciQ · 3 seeds · paired per-(pair,seed) contrasts · pre-registered

Scope & setup

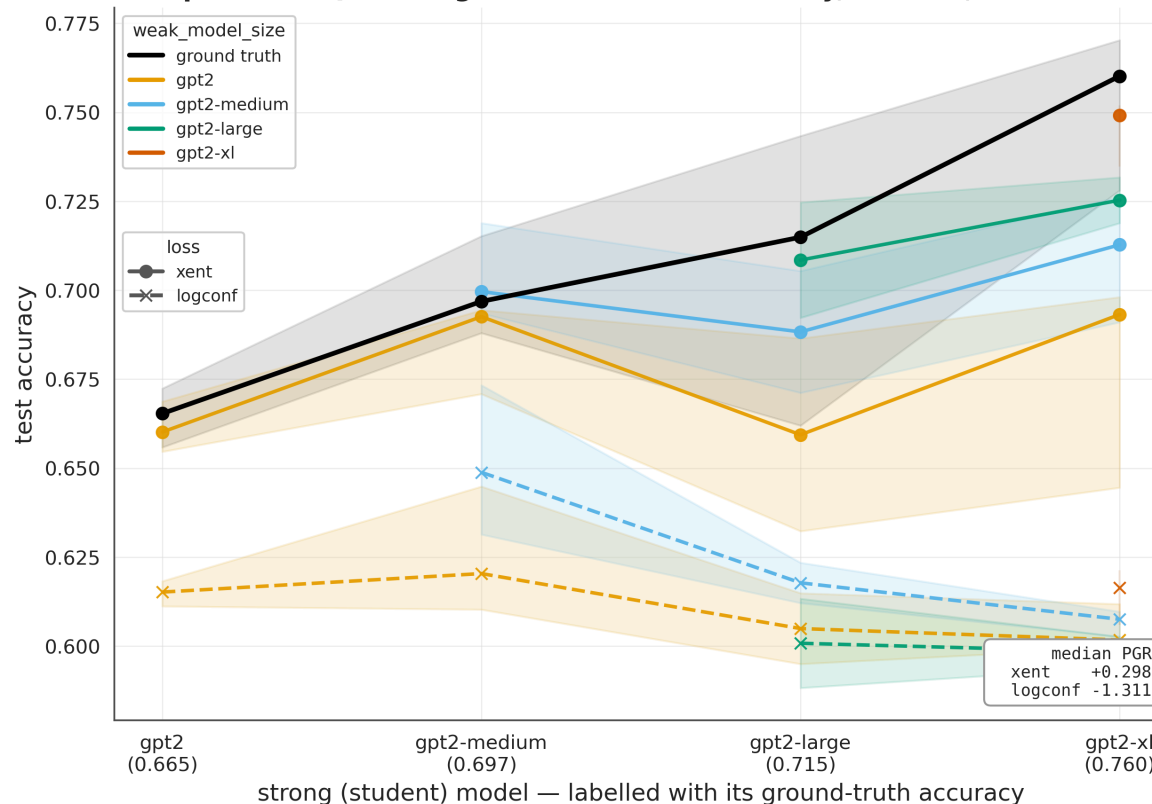
- **Setting:** weak-to-strong generalization — a strong *student* is trained on a weak *teacher's* labels.
- **Extension:** a *supervision budget* — mix a fraction of ground-truth ("strong") labels into the weak labels, and sweep that fraction.
- **Models: GPT-2 family only** — gpt2 / medium / large / xl, with within-family student-teacher pairs.
- **Tasks: BoolQ** (required) + **SciQ** (cross-task check).
- **Readout:** median **PGR** across the model sweep (raw accuracy primary; PGR secondary).

3 seeds · paired per-(pair,seed) contrasts · pre-registered · gpt2-large (seed 1) excluded by rule

GPT-2 sweep at 25% ground truth

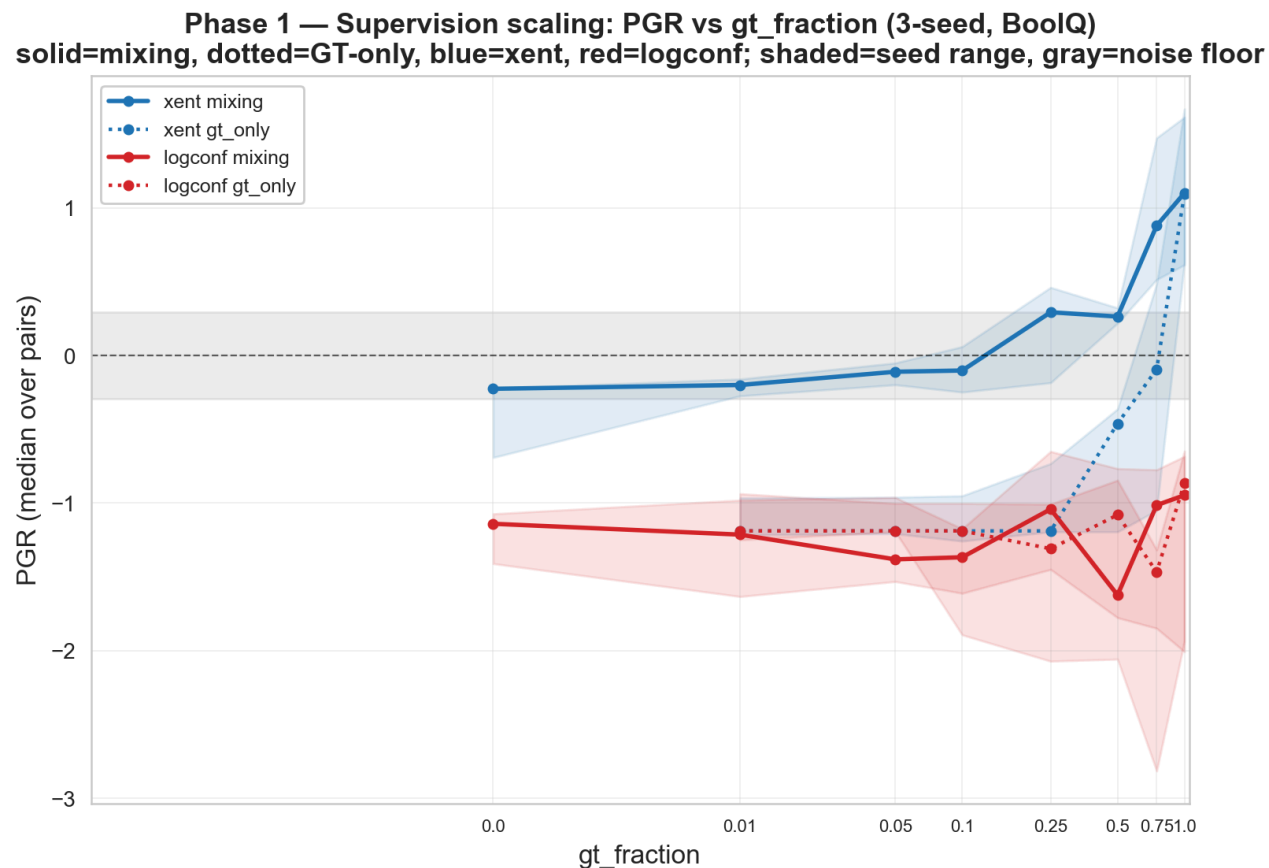
- GPT-2 family, BoolQ, **25% ground truth mixed into the weak labels** — standard sweep format.
- **Median PGR (xent) = +0.30** across the sweep; logconf (dashed) sits well below xent — inert, so we use xent.
- The 0% baseline is **-0.27**; 25% GT moves the median to **+0.30** — a small positive.

W2SG sweep — BOOLQ + 25% ground truth (GPT-2 family, 3 seeds, band = seed range)



How *much*? — back-loaded, and saturating by ~75%

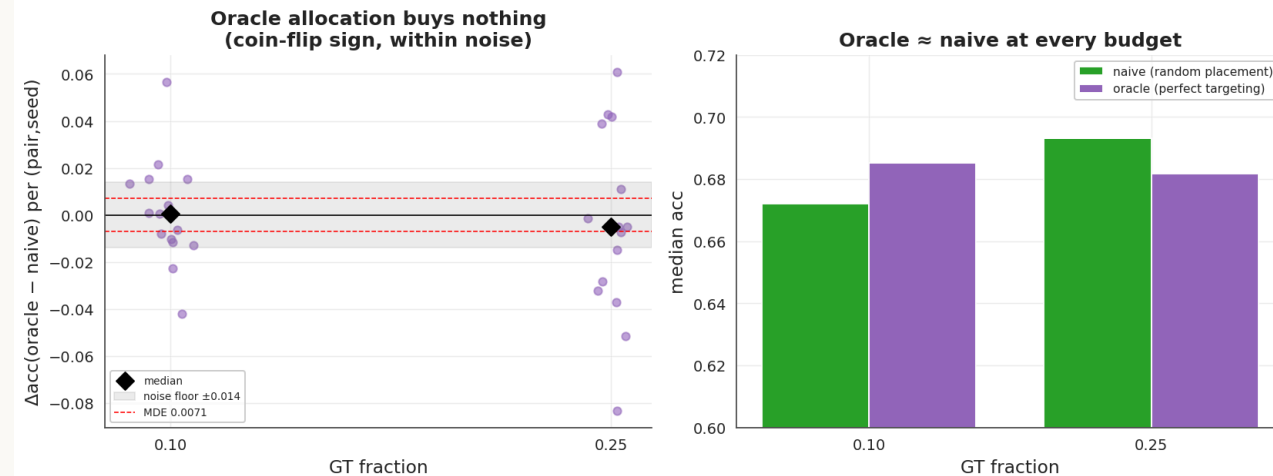
- Up to **10% GT**, the gain over the 0% baseline stays within the **0.014 noise floor**.
- Median xent PGR is **back-loaded**: $-0.22 \rightarrow +0.30$ (0.25) $\rightarrow +0.28$ (0.50) $\rightarrow +0.90$ (0.75) $\rightarrow +1.04$ (1.0).
- The 0.75 point: the **0.50 $\rightarrow$ 0.75 step is the largest**, and **0.75 $\rightarrow$ 1.0 is within noise** — the gain is essentially complete by ~75%.
- A pre-registered concave-knee-at-25% prediction was **refuted**, and retracted after the multi-seed data.



Where it's spent — within noise (allocation null)

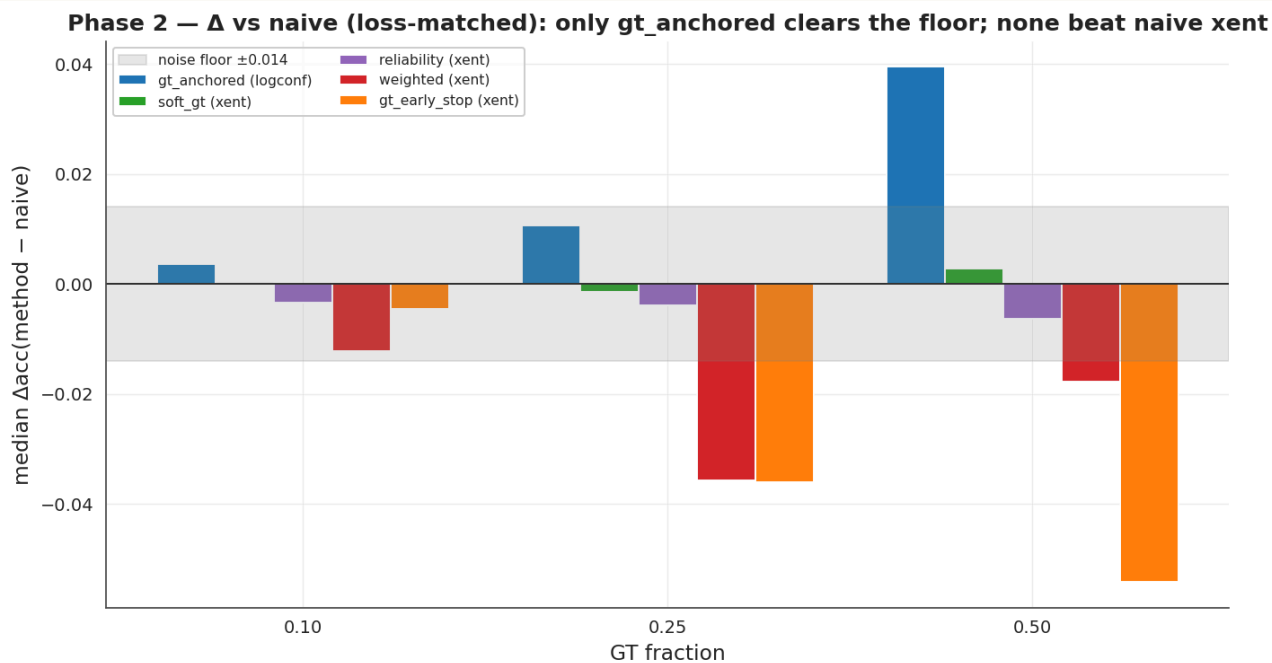
- An **oracle** uses held-out GT to place the budget exactly on the **teacher-wrong rows** — an upper bound on any allocation rule.
- Paired against random placement at matched budget: oracle – random = **+0.0006** (0.10), **-0.0049** (0.25) — both within the **MDE (0.0071)**, so a genuine null, not underpowered.
- If the best possible placement ties random, allocation heuristics have **little to gain** on this testbed.

Phase 1b · B — WHERE you spend GT does not matter (allocation NULL): a perfect error-targeting oracle ties random



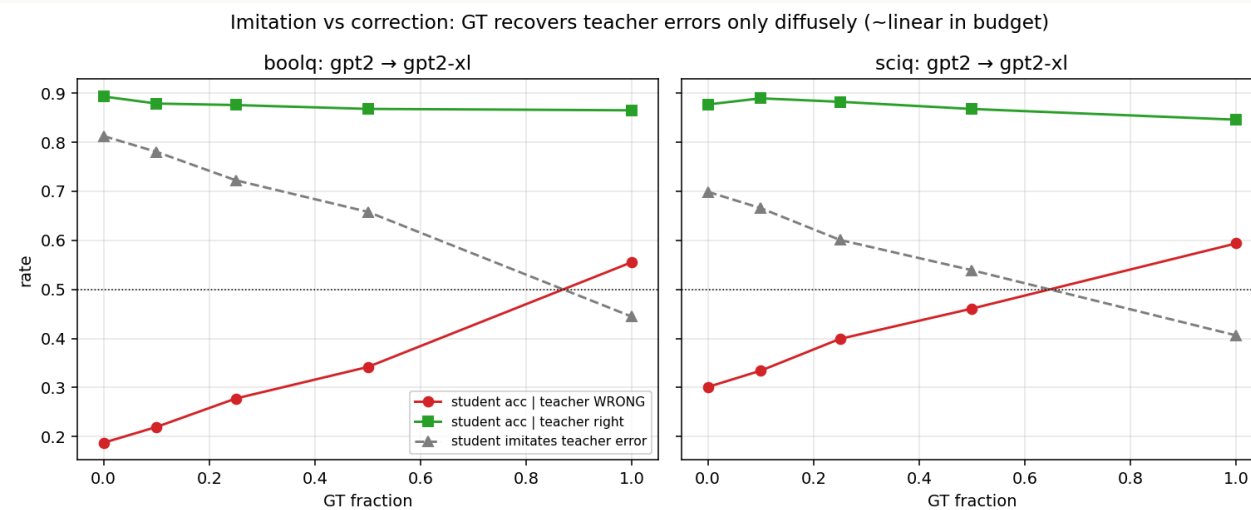
How it's combined — within noise (combination null)

- Five methods vs naive mixing at matched budget: **M1** GT up-weighting · **M2** soft-GT targets · **M3** GT-anchored logconf · **M4** reliability-weighted weak labels · **M5** GT-based early stopping.
- Median Δ vs naive ≈ 0 across $\{0.10, 0.25, 0.50\}$; none shifts the curve left or raises the ceiling.
- **M3 (gt-anchored)** is the only method above the floor (+0.040 at 0.50), but it recovers logconf toward xent **without exceeding plain xent** (0.642 vs 0.697).



Mechanism: the student reproduces the teacher's errors

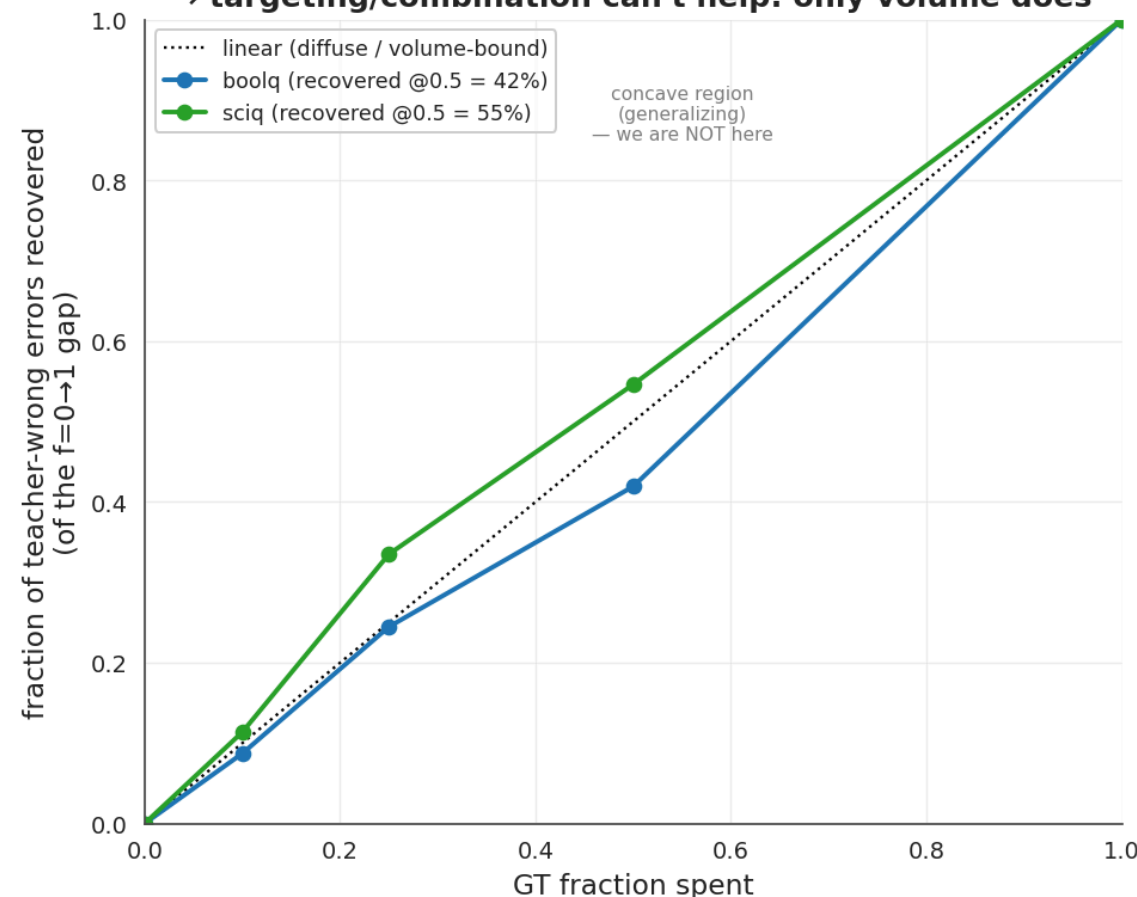
- Probe: gpt2 → gpt2-xl, joining teacher and student **per-example** test predictions.
- On teacher-wrong rows at 0% GT, the student reproduces the teacher's wrong answer **81% (BoolQ) / 70% (SciQ)** of the time.
- Errors are **largely inherited rather than independent** — so the rows needing correction are identifiable in principle.



Mechanism: recovery is ~linear in budget (volume-bound)

- Recovery of teacher-wrong rows is **~linear in budget** — 42% / 55% of the 0→100% gap recovered at 50%, tracking $y = x$ rather than a concave curve.
- A GT label's marginal value appears **roughly independent of which row it lands on**.
- Under volume-bound recovery, placement (*where*) and re-weighting (*how*) have **little leverage** — consistent with both nulls.

Mechanism — error recovery is ~LINEAR in budget (diffuse), not concave
→ targeting/combination can't help: only volume does



Pre-registration scorecard — hits and misses

Phase 1 (P1–P6, git-anchored before seeds 0/2 existed)

	Prediction	Outcome
P1	knee at ~25%	✗ refuted — back-loaded, no knee (retracted)
P2	$\leq 10\%$ GT inert	✓ within noise
P3	mixing > GT-only	✓ (confound named → resolved in 1b)
P4	logconf null	✓ inert at every budget
P5	scale interaction (gap → more GT)	— underpowered at GPT-2 scale
P6	0.25–0.50 plateau	✓ flat in that range

Phase 2 (M1–M5, registered before the portfolio ran): M2 ✓ null, M4 ✓ null (as flagged uncertain), **M3** 🟡 (strongest bet — rescues logconf, still < xent), M1/M5 ✗ negative.

Experiments across the three axes

Axis	Experiment	Outcome
How much	8-point fraction sweep, $0 \rightarrow 1$	back-loaded, saturates ~ 0.75
Where	error-targeting oracle + random control	null – within MDE
How	5 combination / loss methods	null – 1 above floor, still $< \text{xent}$
Loss	xent vs logconf	logconf inert $\rightarrow$ dropped
Tasks	BoolQ + SciQ	replicates on both
Mechanism	imitation-vs-correction probe	recovery $\sim$ linear in budget
Robustness	8-seed variance; 5-seed baseline pass	exclusion by rule; 3-seed conclusions hold

Coverage spans the three axes plus loss, task, and robustness checks; the negatives are concordant with the volume-bound account.

Takeaways

- Reproduced W2SG on the GPT-2 family and extended it to a supervision-budget setting.
- Decomposed the budget question into *how much* / *where* / *how* — a powered null on the last two, and a back-loaded, saturating curve on the first.
- One account — **inherited errors plus volume-bound recovery** — is consistent with all three.
- So the binding constraint here looks like **scale**, not allocation or combination — which sets the next experiments.

Next steps — highest-leverage experiments

1. **Scale the capability gap — the decisive test.** Widen the student-teacher gap (handicapped teachers within GPT-2; larger families beyond it). Volume-bound recovery predicts it stays ~linear; if it turns **concave**, the *where* and *how* axes reopen. Directly falsifies the central claim.
2. **Is teacher error targetable? — closes the allocation null.** Probe whether "teacher-wrong" is predictable from the student's representations. If error is unstructured, that *is* why targeting can't help. Near-zero compute.
3. **The untested axis — *when*.** Schedule the budget (GT-first / anneal / interleave) instead of mixing it statically; iterative-label-refinement results suggest order can matter under weak supervision.
4. **Compounding via iterative weak-to-strong.** Spend the same budget across self-training rounds — does a one-shot-null budget accumulate?

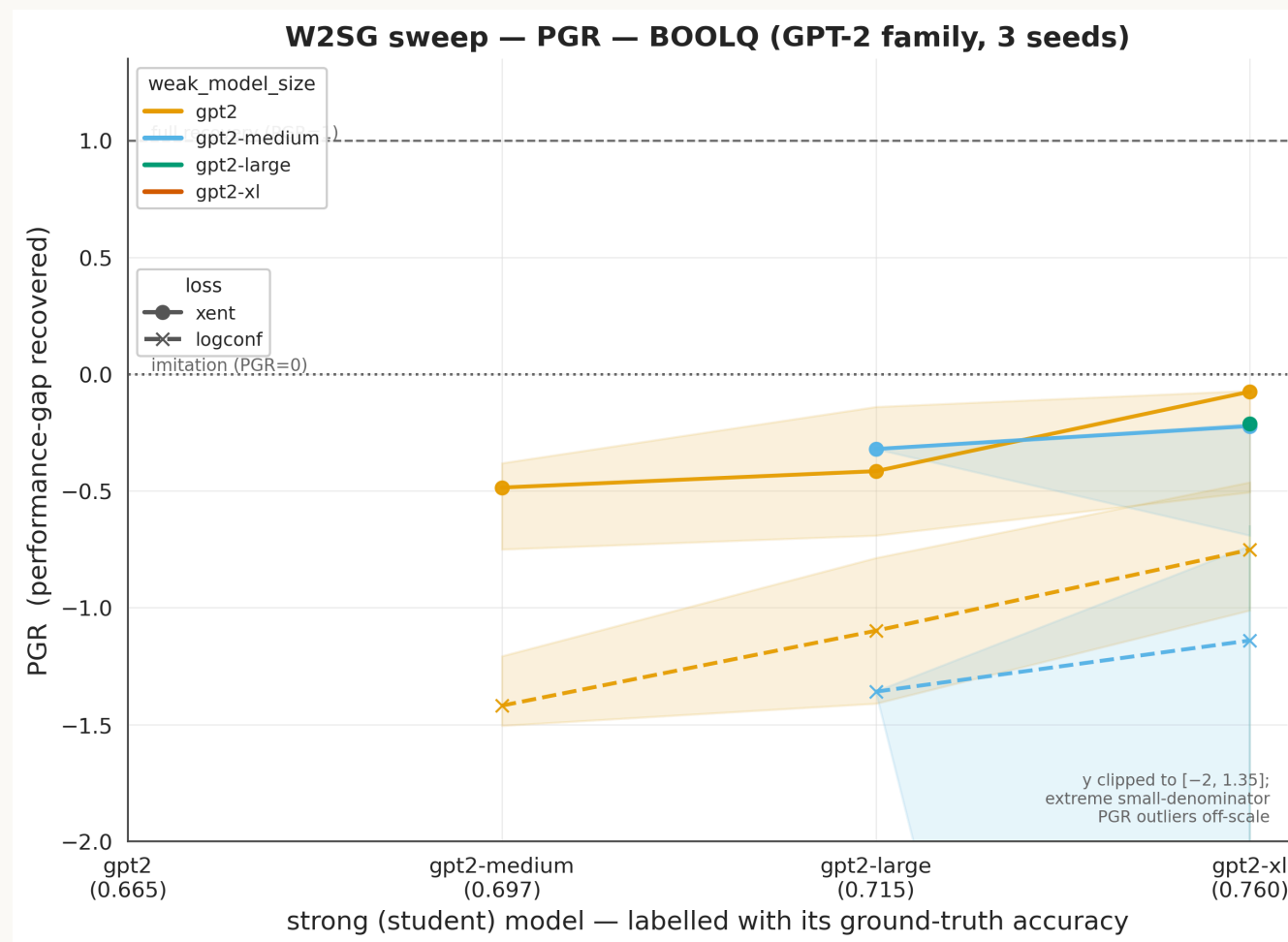
Ordered by leverage: #1 falsifies the central claim · #2 closes the mechanism · #3-4 probe untested axes.

Appendix

frac=0 reproduction (acc + PGR) · transfer heatmap · per-method × pair detail · de-confound · cross-task

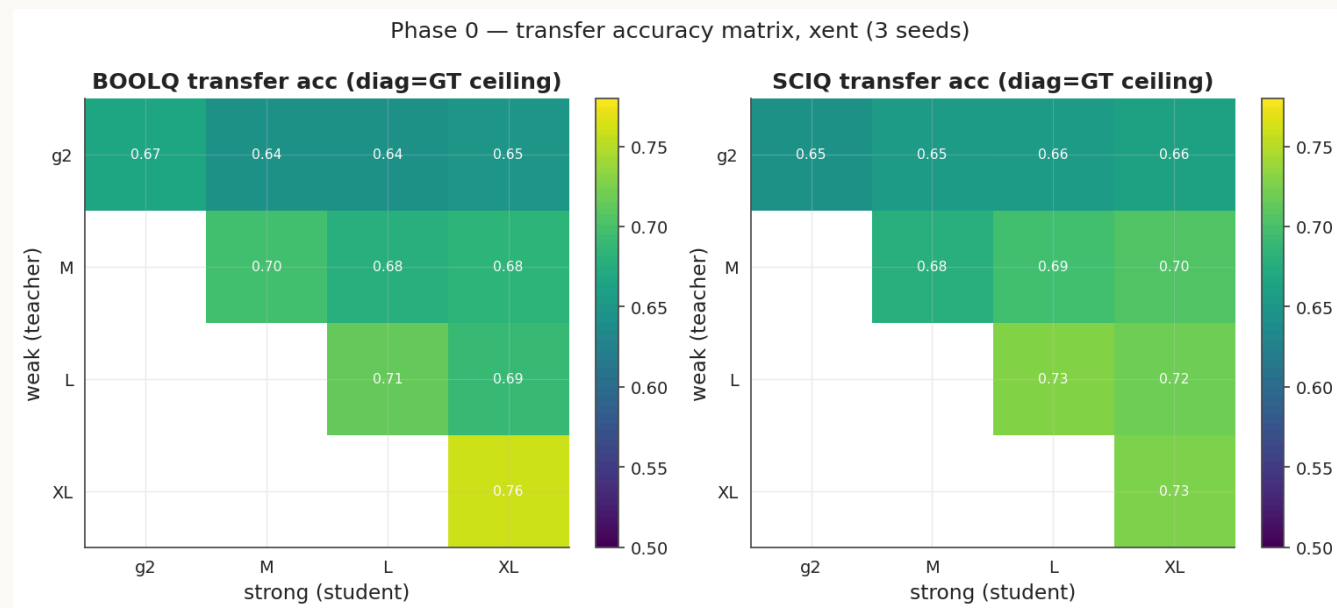
Appendix — frac=0 reproduction

- Standard W2SG sweep, GPT-2 family, BoolQ, **0% GT**, PGR axis.
- Median xent PGR **-0.27**; several pairs fall below the imitation line — BoolQ is low-signal at 0% GT.
- This is the baseline the 25%-GT sweep is measured against.



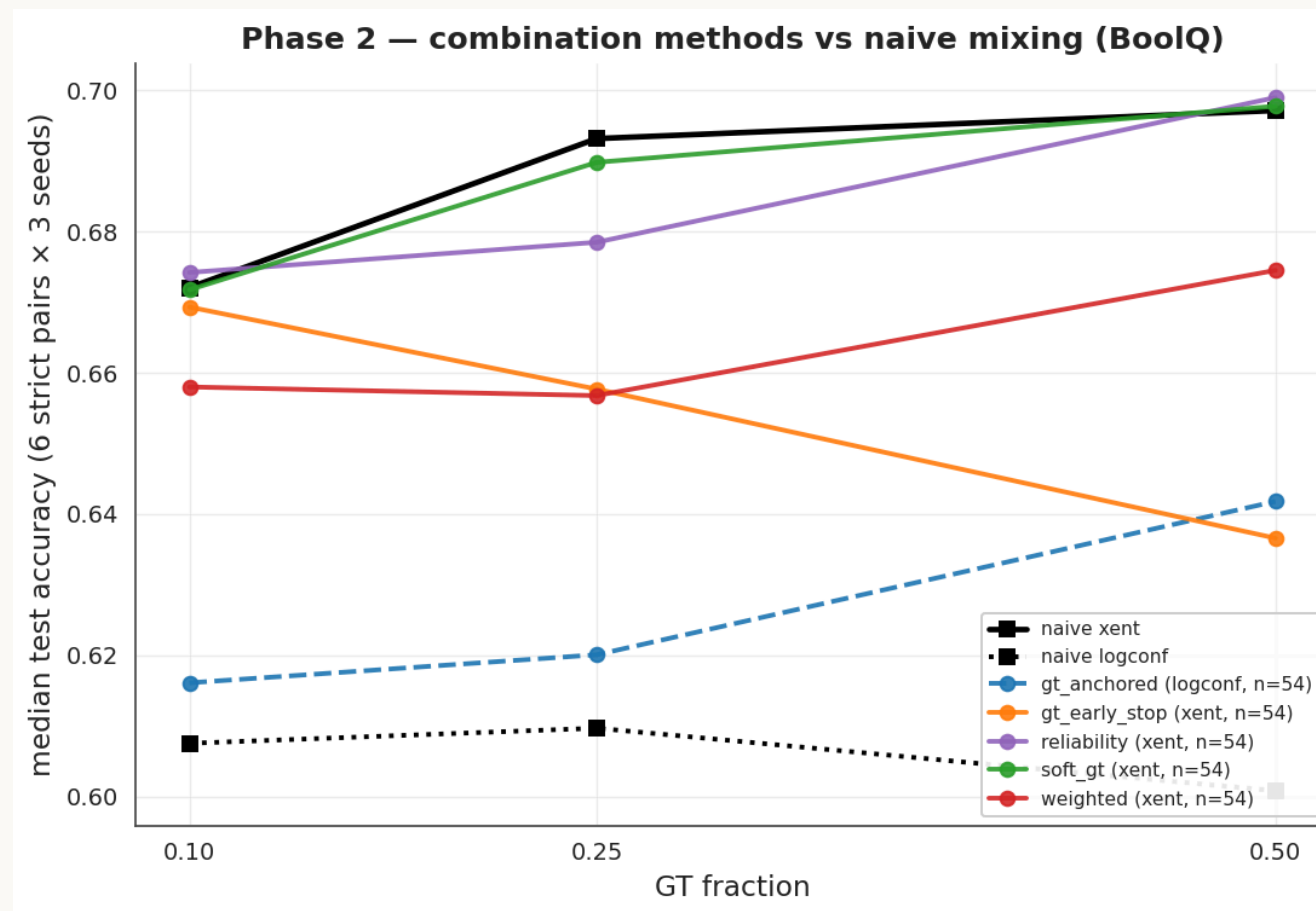
Appendix — transfer tracks the teacher more than student capacity

- With the teacher fixed, scaling the student adds $\sim +0.003$; with the student fixed, a better teacher adds $\sim +0.04$.
- Transfer tracks the teacher far more than student capacity — the static counterpart of the imitation result.



Appendix – combination portfolio, per-method detail

- All five methods track naive mixing across $\{0.10, 0.25, 0.50\}$.
- gt-anchored (logconf) is the only method above the floor; it recovers logconf but stays under xent.



Appendix — weak labels carry information (de-confound)

- Ordering: **random** < **gt_only** < **naive mixing** (15/15 BoolQ, 18/18 SciQ pairs).
- Replacing weak labels with noise reduces accuracy → the mixing gain reflects weak-label information, not just added rows.

